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Third SEMESTER–B.TECH- Course Work
MID-SEMESTER EXAMINATION, September 2026

Course Code-ICMTC301/EEMTC301/CMMTC301
Course Title-Engineering Mathematics
Time: 1:30 hours Max Marks-20

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

<table>
<tr><th>Q. No.</th><th>Question</th><th>Marks</th><th>CO</th></tr>
<tr><td>1 (a)</td><td>Express $f(x) = \frac{x}{2}$ as a Fourier series in the interval $-\pi < x < \pi$.</td><td>2</td><td>CO1</td></tr>
<tr><td>(b)</td><td>Find the half range sine series for $f(x) = e^{ax} - e^{-ax}$ in $0 < x < \pi$.</td><td>2</td><td>CO1</td></tr>
<tr><td>2 (a)</td><td>If $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$, where $f(x) = \begin{cases} \pi x & 0 \le x \le 1 \\ \pi(2-x) & 1 \le x \le 2. \end{cases}$ Find b_n.</td><td>2</td><td>CO1</td></tr>
<tr><td>(b)</td><td>Analyse harmonically the data given below and find a_0 and a_1,
<table>
<tr><td>x</td><td>0</td><td>$\pi/3$</td><td>$2\pi/3$</td><td>π</td><td>$4\pi/3$</td><td>$5\pi/3$</td><td>2π</td></tr>
<tr><td>y</td><td>1.98</td><td>2.15</td><td>2.77</td><td>-0.22</td><td>-0.31</td><td>1.43</td><td>1.98</td></tr>
</table></td><td>2</td><td>CO1</td></tr>
<tr><td>3 (a)</td><td>Find directional derivative of the function $\phi = 4xz^3 - 3x^2yz^2$ at (2,-1,2) along z-axis.</td><td>2</td><td>CO2</td></tr>
<tr><td>(b)</td><td>Determine the constant a so that the vector $\vec{F} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal.</td><td>2</td><td>CO2</td></tr>
<tr><td>4 (a)</td><td>Evaluate $\int_C [(2x^2y + y + z^2)\hat{i} + 2(1+yz^3)\hat{j} + (2z + 3y^2z^2)\hat{k}].d\vec{r}$, along the curve $y^2 + z^2 = a^2$, $x = 0$.</td><td>2</td><td>CO2</td></tr>
</table>

Q. No.	Question	Marks	CO
(b)	Determine whether $\vec{F} = (y^2 \cos x + z^3)\hat{i}$ $+(2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$, is a conservative vector field.	2	CO2
5 (a)	Evaluate by Green's theorem, $\int_C e^{-x}(\sin y\ dx + \cos y\ dy)$ where C is the rectangle with vertices (0,0), $(\pi, \frac{\pi}{2})$, $(0, \frac{\pi}{2})$ and $(\pi, 0)$.	2	CO2
(b)	Evaluate $\iint_S \vec{A}.\hat{n}dS$, where $\vec{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.	2	CO2

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